Quiz 17 - Simple Harmonic Motion

Name: Class:

The number of marks for each question or part question is shown in brackets [ ]. Total mark for this section: 35.

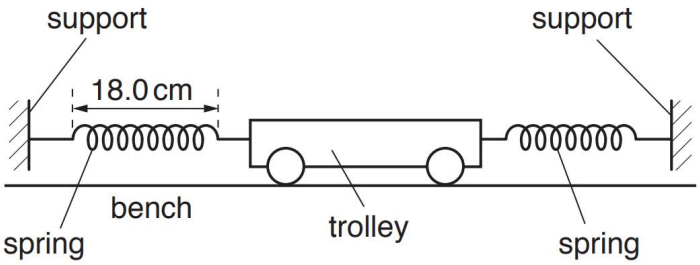
1.

a. According to the defining equation of simple harmonic motion.

State the significance of the minus (–) sign in the equation.

..... [1]

b. A trolley rests on a bench. Two identical stretched springs are attached to the trolley as shown in Fig. The other end of each spring is attached to a fixed support.



The unstretched length of each spring is 12.0 cm. The spring constant of each spring is 8.0 Nm^{-1}. When the trolley is in equilibrium the length of each spring is 18.0 cm.

The trolley is displaced 4.8 cm to one side and then released. Assume that resistive forces on the trolley are negligible.

i. Show that the resultant force on the trolley at the moment of release is 0.77N. [2]

ii. The mass of the trolley is 250g.

Calculate the maximum acceleration a of the trolley.

a = ...ms^{-2} [1]

iii. Use your answer in (ii) to determine the period T of the subsequent oscillation.

T = ...s [3]

iv. The experiment is repeated with an initial displacement of the trolley of 2.4cm. State and explain the effect, if any, this change has on the period of the oscillation of the trolley.

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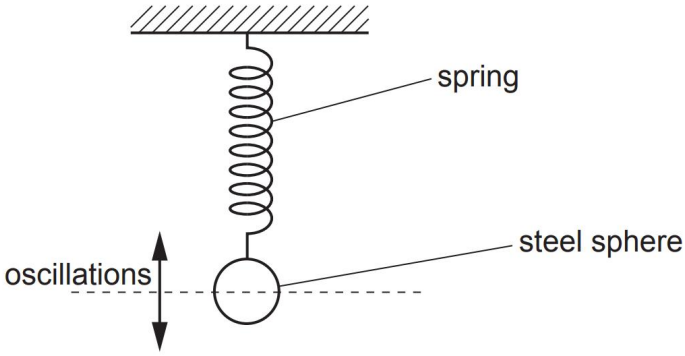
.....[2]

[Total: 9]

2. A small steel sphere is oscillating vertically on the end of a spring, as shown in Fig.



The velocity v of the sphere varies with displacement x from its equilibrium position according to

$$v = \pm\, 9.7\sqrt{(11.6 - x^2)}$$

where v is in cms^{-1} and x is in cm.

a.

i. Calculate the frequency of the oscillations.

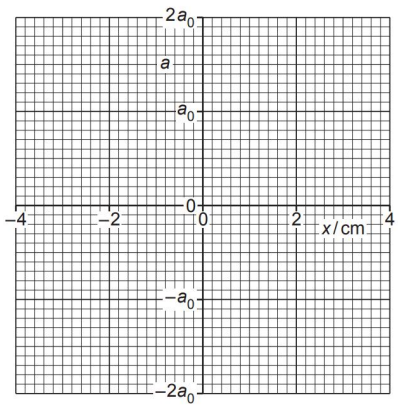
frequency = ...Hz [2]

ii. Show that the amplitude of the oscillations is 3.4cm. [1]

iii. Calculate the maximum acceleration a_0 of the sphere.

a_0 = ...ms^{-2} [2]

b. On Fig., sketch the variation with x of the acceleration a of the sphere.



c. Describe, without calculation, the interchange between the potential energy and the kinetic energy of the oscillations.

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.....

.....[3]

[Total: 11]

3.

a. State what is meant by resonance.

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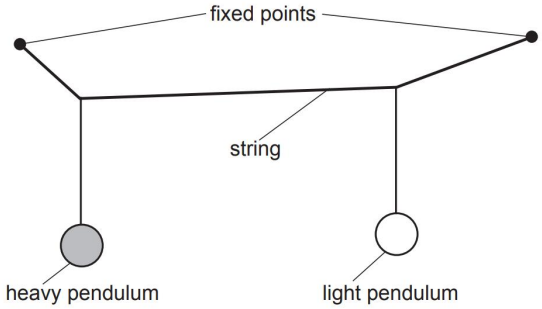
.....[2]

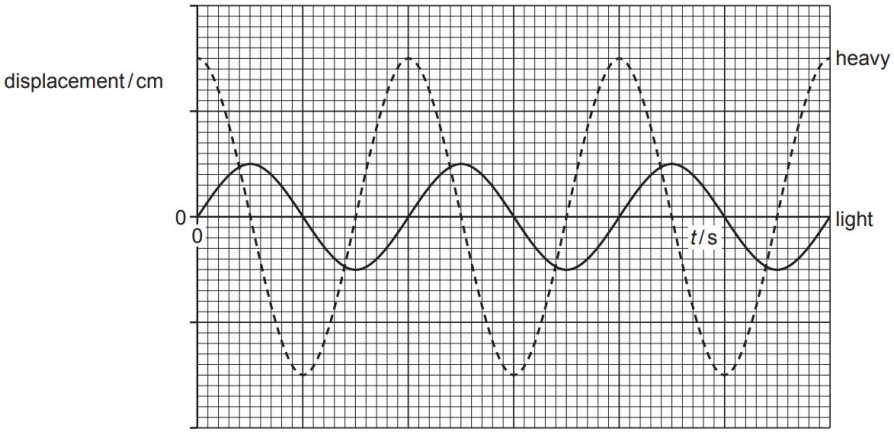
b. Fig. shows a heavy pendulum and a light pendulum, both suspended from the same piece of string. This string is secured at each end to fixed points.



Both pendulums have the same natural frequency.

The heavy pendulum is set oscillating perpendicular to the plane of the diagram. As it oscillates, it causes the light pendulum to oscillate.

Fig. shows the variation with time t of the displacements of the two pendulums for three oscillations.



The variation with t of the displacement x of the light pendulum is given by

$$x = 0.25 sin 5.0\pi t$$

where x is in centimetres and t is in seconds.

i. Calculate the period T of the oscillations.

T = ...s [2]

ii. On Fig. 4.2, label both of the axes with the correct scales. Use the space below for any additional working that you need. [2]

iii. Determine the magnitude of the phase difference ϕ between the oscillations of the light and heavy pendulums. Give a unit with your answer.

ϕ = ... unit ..... [2]

[Total: 8]

4.

a. Explain what is meant by the natural frequency and forced frequency of vibration of a system.

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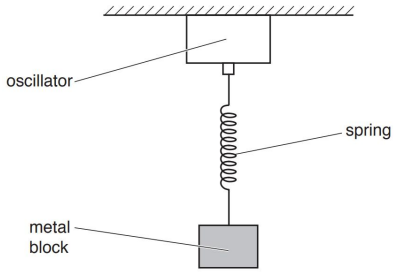
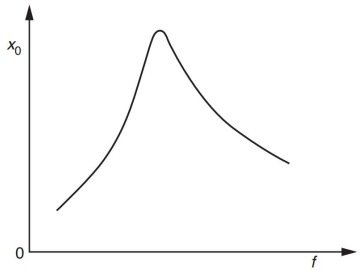
.....[2]

b. A block of metal is fixed to one end of a vertical spring. The other end of the spring is attached to an oscillator, as shown in Fig.



The amplitude of oscillation of the oscillator is constant.

The variation of the amplitude x_0 of the oscillations of the block with frequency f of the oscillations is shown in Fig.



i. Name the effect shown in Fig.

.....[1]

ii. State and explain whether the block is undergoing damped oscillations.

.....

.....[2]

iii. State two examples in which the effect shown in Fig. is useful.

1.....

2.....[2]

[Total: 7]